

INFORMATICS INSTITUTE OF TECHNOLOGY

In Collaboration with

UNIVERSITY OF WESTMINSTER



**MoodMirror: Adaptive Machine Learning System for Personalized
Mental Health Indicators Detection in Social Media Behavior**

Final Year Project by
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Presented in partial fulfilment of the requirements for the BSc (Hons)
Computer Science degree at the University of Westminster.

EXPERT VALIDATION SIGNOFF & CONFIDENTIALITY COMMITMENT LETTER

MOODMIRROR PROJECT SUMMARY

Expert Validation Package

PROJECT OVERVIEW

MoodMirror is a machine learning system designed to identify users at risk of mental health crises by analyzing behavioral patterns from publicly available Reddit posts. The system aims to support early intervention by classifying mental health risk into two categories: At-Risk (moderate concerns) and Severe Risk (acute crisis indicators).

RESEARCH OBJECTIVE

This validation study measures the performance and reliability of the automated MoodMirror classification system by comparing its predictions against independent expert clinical judgment. Your independent assessment of 10 de-identified cases provides ground truth data for evaluating system accuracy and validating its clinical utility.

VALIDATION CASES

You will review 10 de-identified cases, each containing:

1. Behavioral Summary – Quantified metrics (posting frequency, sentiment, engagement in mental health forums)
2. Sample Posts – Up to 8 clinically relevant posts from the user's history, shown chronologically

DATA CHARACTERISTICS

Source: Publicly available Reddit posts
Anonymization: All personal identifiers removed; cases identified by case ID only
Diversity: 5 At-Risk cases and 5 Severe Risk cases, randomly ordered
Review time: Approximately 3–5 minutes per case

WHY EXPERT VALIDATION MATTERS

Your independent assessment helps establish:

- Ground Truth – Accurate baseline for system evaluation
- Reliability – Inter-rater agreement metrics for publication
- Clinical Credibility – Validation by qualified experts
- Research Rigor – Essential for conference/journal publication

DATA USAGE & CONFIDENTIALITY

EXPERT IDENTITY PROTECTION:

- Your name, medical license, and institution will NOT be published or disclosed in any form.
- Only your clinical classifications and agreement metrics will be reported anonymously.
- No personal or identifiable information about you will appear in any academic output.

PERMITTED USE:

This expert validation will be used STRICTLY for:

- Academic thesis/dissertation
- Peer-reviewed journal publications
- Academic conference presentations
- Research methodology documentation

PROHIBITED USE:

- This validation will NOT be used for:
- Commercial purposes or profit
- Marketing or advertising
- Sharing with third parties without explicit written consent
- Any purpose other than stated academic research

CONFIDENTIALITY STANDARDS:

- All data will be stored securely and accessed only by the research team
- Patient data remains anonymized throughout all research phases
- Findings will be reported in aggregate form only
- No individual cases will be identifiable in any publication

CONSENT FOR PUBLICATION:

I understand that results from this validation study may be published in:

- Peer-reviewed journals
- Academic conferences
- University thesis/dissertation
- Research repositories

In such publications, the expert's identity will be protected, and only anonymized findings will be shared.

RESEARCHER COMMITMENT & SIGNATURE

As the principal researcher, I commit to the following:

- I will maintain strict confidentiality of all expert identifying information
- I will never publish or disclose the expert's name, license, or institution
- I will report only anonymized classifications and aggregate findings
- I will store all signed agreements securely and separately from published data
- I will comply with all data protection and ethical research standards

Researcher Signature:

Researcher Name: Kavind Pramod

EXPERT CREDENTIALS

Full Name:

Professional Title:

Medical License/ID:

Specialization:

Years of Practice:

Institution/Organization:

I acknowledge that I have independently reviewed and clinically classified 10 de-identified cases using the MoodMirror validation materials provided. I confirm that each classification reflects my independent professional judgment based on the established criteria, without external influence or coercion.

PROFESSIONAL ATTESTATION

I certify and confirm the following statements:

- I reviewed all materials independently without external influence or bias.
- I applied consistent clinical judgment based on the classification criteria provided.
- I understand all user data is de-identified and maintained confidentially.
- I understand research findings may be published in academic venues.

I declare no conflicts of interest related to this research or MoodMirror. I am authorized to provide expert clinical validation in mental health assessment.

Expert Signature:

Name: Dr. Aruna wijayanayaka

Date:

Institutional Seal / Stamp (if applicable):

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By signing this document, I confirm that I have independently reviewed and classified the 10 provided cases in accordance with the stated mental health risk criteria and professional clinical standards. I understand this validation will be used for academic research purposes only and may be included in peer-reviewed publications.